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LLM-Based Adjustment of Short-Term Electricity Demand Forecasts: Direct and Hybrid Approaches

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Abstract

Short-term electricity demand can depart from recurring patterns during holidays, transport disruption, major sporting events and extreme weather, yet relevant information may first become available as text. This dissertation examines whether a large language model (LLM) can convert such information into reliable forecast adjustments and which forecasting decisions, if any, should be delegated to it.

A leakage-safe benchmark compares Seasonal Naive, Ridge and LightGBM models across six UK and Ireland regions and forecast horizons of one to thirty days. A frozen East Midlands experiment evaluates direct LLM percentage adjustments under matched relevant-context, shuffled-context and no-context conditions. A hybrid alternative instead restricts the LLM to extracting event type, dates and geographic applicability, while deterministic rules assess eligibility and historical residuals determine adjustment direction and magnitude.

Direct LLM adjustment increases event MAE from 239.80 to 310.23 MW, showing that access to relevant text does not ensure numerical calibration. The hybrid reduces MAE to 131.32 MW on the same observations and transfers positively across supported holiday settings. On an unseen external electricity system, a frozen scale-normalised Ridge model improves France 2024 MAE by 15.14% across all thirty horizons. However, non-calendar evidence is substantially weaker, and prediction intervals become under-calibrated under system shift. Event-level semantic extraction also substantially reduces LLM inference requirements.

Overall, the findings support a limited division of responsibility: LLMs are useful for translating event text into verifiable structured information, but forecast-adjustment authority should remain with historical evidence and deterministic checks. When evidence is insufficient, unsupported or incompatible with the baseline, the forecast should remain unchanged.

Acknowledgements and GenAI declaration

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Abbreviations

- **CI**: confidence interval
- **EDA**: exploratory data analysis
- **LLM**: large language model
- **MAE**: mean absolute error
- **MAPE**: mean absolute percentage error
- **MASE**: Seasonal-Naive-relative mean absolute error used in this study
- **NESO**: National Energy System Operator
- **NLP**: natural language processing
- **RMSE**: root mean squared error
- **RTE**: Réseau de Transport d'Électricité
- **sMAPE**: symmetric mean absolute percentage error
- **WAPE**: weighted absolute percentage error

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Chapter 1 Introduction

1.1 Background and motivation

Short-term electricity-demand forecasts support system balancing, generation scheduling, procurement and operational risk management. Forecasting models normally exploit historical load, weather and calendar regularities, yet these inputs may be insufficient when demand is altered by an unusual event. Public holidays, industrial action, major sporting events and extreme weather can change travel, working and household behaviour before their effects are visible in recent load observations. Much of the relevant information is initially expressed as text rather than as a stable numerical variable.

Natural language processing (NLP) offers one route for converting such information into forecast features. Bai, Camal and Michiorri (2025), for example, show that news-derived social and economic features can improve regional multi-horizon demand forecasting, although the benefit varies by region and horizon. Large language models (LLMs) may interpret event type, timing and geographic relevance more flexibly than fixed word counts or topics. However, this semantic capability does not imply that an LLM can reliably choose the direction or magnitude of a load adjustment. Direct numerical generation may also add cost, model dependence and weak reproducibility.

This dissertation evaluates two ways of using event text to adjust an existing demand forecast. In the direct approach, an LLM receives the eligible event description and selects a bounded percentage adjustment. Relevant-context, shuffled-context and no-context conditions are evaluated on identical forecast rows to determine whether the text itself provides useful information. In the hybrid approach, the LLM extracts event type, dates and geographic applicability, but the numerical adjustment is estimated from historical forecast residuals and applied only after deterministic eligibility checks. The selected methods are subsequently evaluated under regional, temporal, baseline and electricity-system transfer. This design separates three questions: whether the text is relevant, whether its numerical effect can be calibrated, and whether the resulting forecast decision remains useful outside the initial experiment.

1.2 Research gap

Energy-forecasting research establishes the need for chronological baselines and reproducible comparisons (Hong et al., 2020), while news-based studies show that conventional NLP features can add social context (Bai, Camal and Michiorri, 2024; 2025). Context-aware forecasting also shows that language can matter but that pretrained language components do not automatically improve numerical predictions (Williams et al., 2025; Tan et al., 2024). These studies do not establish that an LLM should directly alter an operational load forecast or that a text-driven adjustment transfers to a held-out region, later year, different baseline or new electricity system. The gap is therefore whether explicit event text can produce an accurate, conservative and transferable forecast action.

1.3 Aim, objectives and research questions

The overarching aim of this dissertation is to evaluate the extent to which time-safe, event-related textual context can improve short-term regional electricity-demand forecasting, and to establish an empirically defensible division of responsibility between LLM-based semantic interpretation and conventional numerical forecasting.

The objectives are to:

1. establish leakage-safe structured-data and conventional NLP benchmarks across six regions and forecast horizons of one to thirty days;

2. determine whether relevant event text improves direct LLM forecast adjustments relative to matched shuffled-context and no-context controls;
3. evaluate whether LLM-based semantic extraction combined with historically calibrated adjustments provides a more reliable alternative to direct LLM numerical adjustment; and
4. assess the external validity and operational viability of the hybrid framework beyond its development setting.

In this study, operational viability is evaluated through prediction-interval calibration, abstention behaviour and LLM inference cost.

The research questions are:

1. How much predictive value do structured-data, economic and conventional NLP features provide across regions and forecast horizons?
2. Does relevant event text enable direct LLM adjustment to outperform the underlying baseline and the matched control conditions?
3. Can semantic extraction combined with historical calibration provide a more reliable forecasting mechanism than direct LLM numerical adjustment?
4. To what extent does the hybrid framework remain robust and operationally viable when transferred to unseen regions, time periods and electricity systems?

1.4 Contribution and scope

Existing studies have primarily incorporated aggregated textual features into conventional forecasting models (Bai, Camal and Michiorri, 2024; 2025) or used LLMs as numerical forecasters and forecasting agents (Gruver et al., 2023; Zuo, Qin and Wang, 2025; Wang et al., 2026). However, the literature reviewed in this dissertation provides limited evidence on whether event text itself improves an existing load forecast under matched experimental conditions, whether an LLM should be given forecast-adjustment authority, and whether any resulting mechanism transfers beyond the region and electricity system in which it was developed.

This dissertation makes three contributions. First, it isolates the incremental value of event text by comparing relevant-context, shuffled-context and no-context conditions using identical forecast rows and the same underlying baseline. The resulting negative finding demonstrates that consistent and well-structured LLM responses do not necessarily produce numerically calibrated forecast adjustments. Second, it develops and evaluates a modular hybrid framework that separates semantic interpretation from numerical calibration: the LLM identifies event characteristics, historical residuals determine adjustment magnitude, and deterministic rules allow the system to preserve the baseline when evidence is insufficient. Third, it evaluates this framework beyond a single in-sample setting through strict leave-one-region transfer, later-year forecasting, non-calendar events, alternative baselines and an external French electricity system. Prediction-interval calibration, abstention behaviour and inference cost are also examined to distinguish predictive improvement from practical deployment readiness.

The primary data span June 2020 to mid-2023 and cover six UK and Ireland regions. Conventional text variables consist of aggregated features from the *News and Load* release, while explicit event descriptions are obtained from a separate source-linked catalogue containing publication and operational-availability timestamps. France 2024 provides an external point-forecast evaluation. The study does not claim that an LLM can replace a forecaster, that every textual event has a stable effect on electricity demand, or that the reported uncertainty intervals will remain calibrated in every electricity system.

This dissertation focuses exclusively on short-term regional electricity-demand forecasting, operationalised as forecast horizons of one to thirty days ahead. This forecasting task is distinct from long-term capacity, infrastructure or investment planning. Throughout the dissertation, demand refers to the regional load

variable supplied in the *News and Load* release; the study does not construct a separate net-demand series by subtracting renewable generation.

1.5 Dissertation structure

Chapter 2 reviews forecasting, textual information and LLM-based context use. Chapter 3 describes the data and exploratory audit. Chapter 4 presents the chronological design and the progression from direct adjustment to gated historical calibration. Chapter 5 reports the benchmark, controlled failure, redesign and generalisation evidence. Chapter 6 interprets the mechanisms and claim boundaries, and Chapter 7 concludes.

Chapter 2 Literature Review

2.1 Forecasting foundations and external drivers

Short-horizon demand forecasts support balancing and dispatch, while multi-day forecasts inform procurement and risk preparation. Hong et al. (2020) emphasise reproducible benchmarks across model classes. The same principle applies to text: added model capacity must not be confused with added information.

Load has strong intraday, weekly and seasonal structure, while temperature, calendar position and economic conditions alter demand. Energy benchmarks therefore treat exogenous variables as central (Wang et al., 2023). The post-COVID competition shows how abrupt behavioural change weakens earlier relationships (Farrokhabadi et al., 2022), a risk also relevant to industrial action and exceptional weather.

Forecasting performance depends on the forecast horizon, model class and evaluation design (Hong et al., 2020; Wang et al., 2023). A Seasonal Naive benchmark can be difficult to outperform when weekly demand patterns are stable and provides an interpretable test of whether additional model complexity is useful. Regularised linear models examine whether predictors contribute approximately additive information, while boosted trees allow nonlinear relationships and interactions. Ensembles may reduce model-specific variation, although validation-selected weights can still transfer poorly after a change in regime. These considerations support evaluating contextual information across several established model classes and retaining a horizon-specific fallback rather than attributing the result to one preferred algorithm.

Within the short-term one-to-thirty-day forecasting window examined in this dissertation, the later horizons create an additional information problem. Recent load is highly informative at the one-day horizon but becomes less decisive towards the thirty-day horizon, where known calendars, weather outlooks and social context may carry relatively more information. Horizon-wise evaluation is therefore essential. An average across all thirty horizons can conceal a feature that is useful only after the first week or an adjustment that improves the earliest horizons but decays incorrectly. The regional and multi-horizon structure used by Bai, Camal and Michiorri (2025) is particularly useful because it exposes both forms of heterogeneity.

This motivates contextual forecasting, but it also creates an availability problem. A statutory holiday is known in advance as structured data, whereas a strike or emergency warning may first appear in a dated announcement. A forecasting experiment must distinguish information available at the forecast origin from later reporting. Otherwise, apparently useful text can leak knowledge of the target period. A practical example is provided by the additional bank holiday for the Coronation of King Charles III. The event occurred on 8 May 2023 but was announced by the UK Government in November 2022, so the information was available before every forecast origin used in the experiment (GOV.UK, 2022). By contrast, the UK Health Security Agency (UKHSA) cold-weather alert covering 6–9 March 2023 was updated during the event period (UK Health Security Agency, 2023). Forecasts were permitted to use the alert only from the availability timestamp recorded in the study's event catalogue. This restriction ensures that contextual information is evaluated according to what was known at the forecast origin rather than what became known during or after the target period. Evaluation must also be scale- and use-aware: MAE and RMSE measure different aspects of error, regional comparisons need normalisation, and conventional loss reductions do not necessarily translate directly into operational savings (Zhang, Wang and Hug, 2022).

2.2 Text and conventional NLP in energy forecasting

Text can describe mechanisms that are difficult to encode manually, including event type, expected duration, affected location, behavioural response and uncertainty. Bai, Camal and Michiorri (2024) transform news into word-frequency, sentiment, topic and embedding features for day-ahead UK electricity-demand forecasting and report gains over an official benchmark. Their earlier LSTM study also finds value from

news-derived variables within a sequential model (Bai, Camal and Michiorri, 2023). These findings establish the potential value of text, but they do not demonstrate that its contribution is stable across regions, forecast horizons or non-pandemic periods.

Evidence beyond the *News and Load* studies broadens this picture. Lu, Liu and Wang (2024) construct a text-derived electricity policy uncertainty index and show that its inclusion improves mixed-frequency electricity-demand forecasting. Huang et al. (2025) use natural language processing to construct event-related variables from social-media data for modelling local electricity consumption. These studies demonstrate alternative ways of converting unstructured text into structured demand predictors, but neither evaluates whether dated event text should trigger an adjustment to an existing short-term regional forecast under matched context controls.

The most direct empirical anchor is the regional *News and Load* study (Bai, Camal and Michiorri, 2025). It combines load, economic indicators and news-derived features across UK and Ireland regions and horizons from one to thirty days. Textual and economic factors improve deterministic and probabilistic forecasts on average, yet usefulness varies materially by region and horizon. This heterogeneity has two implications. First, the appropriate comparator is a region- and horizon-specific validated baseline, not one global text-versus-no-text mean. Second, a deployment mechanism needs a fallback because contextual features can add noise or duplicate information already captured elsewhere.

Conventional NLP features can be cached and passed to standard models, but a topic score or embedding does not explain why an event applies to a date or region. High-dimensional news also risks spurious association, making chronological ablation and event-level analysis preferable to an uncontrolled feature block.

Recurring context and unstructured events require different treatment. Weekends and statutory holidays should enter directly as structured calendar variables; LLM use is more defensible when timing, geography or mechanism is not represented by the feature schema. Even then, association is not causality: a news feature may proxy season or policy, and a plausible event does not reveal its counterfactual residual relative to a particular baseline. The present design therefore evaluates paired forecast errors from source-linked events rather than treating feature importance as proof of a social mechanism.

Adjacent electricity-price work reaches a similarly cautious conclusion. Bi et al. (2025) introduce a news-augmented benchmark for LLM price forecasting and report both limited news benefits and malformed numerical outputs. Price and load are different targets, but the numerical-output problem is relevant: linguistic coherence does not guarantee calibrated quantities.

2.3 LLMs, time series and context

One forecasting strand represents numerical sequences with language-model architectures: LLMTime encodes observations as strings (Gruver et al., 2023), Time-LLM reprograms a frozen model with time-series patches (Jin et al., 2024), and Chronos tokenises scaled values (Ansari et al., 2024). These methods show representational feasibility but do not justify replacing a strong task-specific forecaster.

Critical work reinforces that distinction. The survey by Zhang et al. (2024) identifies a continuing modality gap between language and numbers. Tan et al. (2024) show through ablation that some apparent LLM forecasting gains do not depend on the pretrained language component. A defensible study must therefore isolate what natural-language capability adds, rather than attaching an LLM label to a wider model change.

The strongest rationale arises when text contains information unavailable in the numerical history. The *Context is Key* benchmark pairs time series with natural-language context required for prediction and shows that multimodal reasoning can be necessary (Williams et al., 2025). Its limitation is external validity: the benchmark spans many domains and includes designed scenarios rather than a noisy operational news stream. The benchmark therefore shows that natural-language context can be necessary in controlled

forecasting examples, but it does not establish that dated event text improves regional electricity-demand forecasts under operational conditions.

Reproducibility is another dividing line. Proprietary LLM behaviour depends on a dated model snapshot, prompt, schema and provider implementation; temperature zero does not guarantee calibrated output. Stored requests, responses and hashes document the tested mechanism but cannot make one model representative of all LLMs.

Energy-specific LLM frameworks have begun to address this gap. Zuo, Qin and Wang (2025) position an LLM as an interactive coordinator that can incorporate operator knowledge. Wang et al. (2026) combine regional news analysis, an LLM agent and direct next-day load generation. These systems broaden the role of language models, but their coupled pipelines make it difficult to isolate whether gains arise from source selection, semantics, fine-tuning or the numerical generator.

2.4 From direct adjustment to a hybrid gate

Three roles for LLMs can be distinguished in the existing forecasting literature. First, an LLM may generate the numerical forecast directly, either by treating the time series as a sequence or as part of a news-driven forecasting agent (Gruver et al., 2023; Wang et al., 2026). Second, an LLM-derived representation may be incorporated into a wider forecasting architecture (Jin et al., 2024; Williams et al., 2025). Third, the LLM may act as a semantic or interactive component that converts contextual information into structured inputs for a separate forecasting model (Zuo, Qin and Wang, 2025). The third role provides the clearest audit trail for the present study. Event type, dates and geographic coverage can be checked against the source, while adjustment direction and magnitude can be estimated from historical residuals. It also permits abstention: if the text is unavailable at the forecast origin, geographically irrelevant or unsupported by calibration data, the baseline remains unchanged.

This separation distinguishes semantic uncertainty about whether an event applies from numerical uncertainty about whether comparable events show a stable residual. Prediction intervals quantify residual dispersion, whereas abstention asks whether the evidence justifies acting at all. Both can fail under shift: an event factor may double-count context already embedded in a national baseline, and an internally calibrated interval may under-cover in a new electricity system. Strict target-region exclusion, later-year tests and baseline-compatibility checks are therefore required before calling the mechanism transferable.

2.5 Research position and testable propositions

The unresolved question is not whether news contains social information; conventional NLP evidence already supports that claim. It is whether explicit, time-safe event text can produce a reliable forecast action beyond a strong baseline, and whether that action remains conservative under distribution shift. This distinction separates semantic information from forecast-adjustment authority: recognising that an event is relevant is not equivalent to knowing the counterfactual residual that should be added to an existing forecast.

Four testable propositions follow from the limitations identified in the literature.

P1: The incremental value of contextual predictors will vary across regions and forecast horizons. Bai, Camal and Michiorri (2024; 2025) report that the contribution of news-derived variables is heterogeneous rather than uniform. A single pooled text-versus-no-text comparison could therefore conceal both useful and harmful regional or horizon-specific effects. This motivates region- and horizon-specific validation against several conventional forecasting models.

P2: A direct LLM adjustment can be structurally consistent without improving forecast accuracy. Research on the modality gap between language and numerical reasoning, together with evidence of malformed or poorly calibrated numerical outputs, shows that coherent text generation does not guarantee accurate quantities (Zhang et al., 2024; Tan et al., 2024; Bi et al., 2025). This motivates the matched relevant-

context, shuffled-context and no-context experiment, which tests whether event text improves the adjustment rather than merely changing the model response.

P3: Separating semantic interpretation from adjustment calibration should provide a more reliable mechanism for supported event families. Event type, dates and geographic applicability can be checked against a source, whereas the demand effect of an event depends on the underlying baseline and must be estimated empirically. The hybrid architecture therefore assigns semantic extraction to the LLM, eligibility decisions to deterministic rules and adjustment magnitude to historical residuals. This proposition is evaluated only where comparable historical events are available.

P4: Forecast improvements and uncertainty calibration will not necessarily transfer unchanged across regions, years, baselines or electricity systems. Regional demand structure, baseline construction and residual distributions can change under transfer, while conformal coverage can deteriorate when calibration and evaluation observations are not exchangeable (Angelopoulos and Bates, 2023). This motivates strict target-region exclusion, later-year and external-system tests, baseline-compatibility checks, prediction intervals and an explicit option to preserve the baseline.

These propositions determine the order of the empirical analysis. The conventional benchmark evaluates P1, the controlled direct-adjustment experiment evaluates P2, the hybrid redesign evaluates P3, and the generalisation and uncertainty experiments evaluate P4. They are testable expectations rather than assumptions of success: each can be rejected, supported only for particular event families or narrowed by the external evidence.

Chapter 3 Data and Exploratory Analysis

3.1 Data source and empirical scope

The empirical analysis uses the complete local release associated with the *News and Load* regional multi-horizon forecasting study (Bai, Camal and Michiorri, 2025). The release combines regional electricity demand, hourly temperature, calendar indicators, economic variables and daily news-derived features. Six areas are retained: East Midlands, West Midlands, South West, South Wales, Northern Ireland and Ireland. For every region, the release provides 30 forecast-horizon files covering one to thirty days ahead.

The canonical series spans 1 June 2020 to 30 June 2023 and contains 1,125 UTC-labelled days per region, each stored with 24 hourly slots. Across six regions this gives 162,000 potential demand observations. All 180 expected region-horizon files are present, with unique forecast-origin indices and the required 24-column target schema.

3.2 Forecast alignment and study partitions

A row is indexed by its forecast origin. For horizon h, the target date is forecast origin + h days. This distinction is important: variables available at the origin must not be joined using the future target date. For the one-day file, the 24 target values exactly reproduce the following day's 24 load values in every finite cell for all six regions; the maximum absolute difference is zero. The H=1 input load matrix can therefore be used as the non-duplicated descriptive time series, while the horizon-specific files remain the modelling inputs.

Formal modelling reindexes rows by target date: training precedes 1 January 2022, validation covers 2022, and testing begins on 1 January 2023. Counts vary slightly by horizon and missingness, and the longest horizons extend into July 2023. Selection preprocessing uses training only; after parameters are frozen, the final test fit may use training plus validation. No test metric selects a model.

3.3 Data quality, missingness and time representation

Only 822 of 162,000 demand cells are missing (0.507%), and all observed values are positive. Missing values are imputed by fitting-partition medians after the chronological split. Table 3.1 reports scale and missingness.

Table 3.1. Regional demand scale and missingness.

Region	Mean demand (MW)	99th percentile (MW)	Demand missing (%)	Temperature missing (%)
East Midlands	2874.5	4156.8	0.144	0.089
West Midlands	2728.7	3834.0	0.144	0.089
South West	1368.2	2084.5	0.167	0.089
South Wales	1216.0	1692.2	0.167	0.089
Northern Ireland	855.0	1328.6	1.741	0.089
Ireland	3549.4	4899.1	0.681	0.089

The fixed 24-column UTC representation contains no 23- or 25-hour local days. The calendar is contiguous and unique; daylight-saving transitions are normalised rather than encoded as variable-length days, so columns must not be interpreted as local clock hours.

The released holiday flags require correction. They are object-valued strings, are identical across all six regions, and contain no flagged day in 2023. This is incompatible with the different statutory calendars of England and Wales, Northern Ireland and Ireland. The modelling experiments therefore rebuild numerical holiday features by target date and jurisdiction from a frozen official calendar snapshot.

3.4 Regional demand patterns

Figure 3.1 shows that absolute load scale differs substantially across regions, so raw MAE should not be used for cross-region ranking without a normalised measure such as MAPE, WAPE or MASE. Figure 3.2 divides each observation by its regional mean and reveals the common daily structure used by the conventional forecasting baselines. The intraday profiles show a low-demand overnight period, a morning increase and an evening peak. The magnitude and timing of these patterns are nevertheless region-specific.

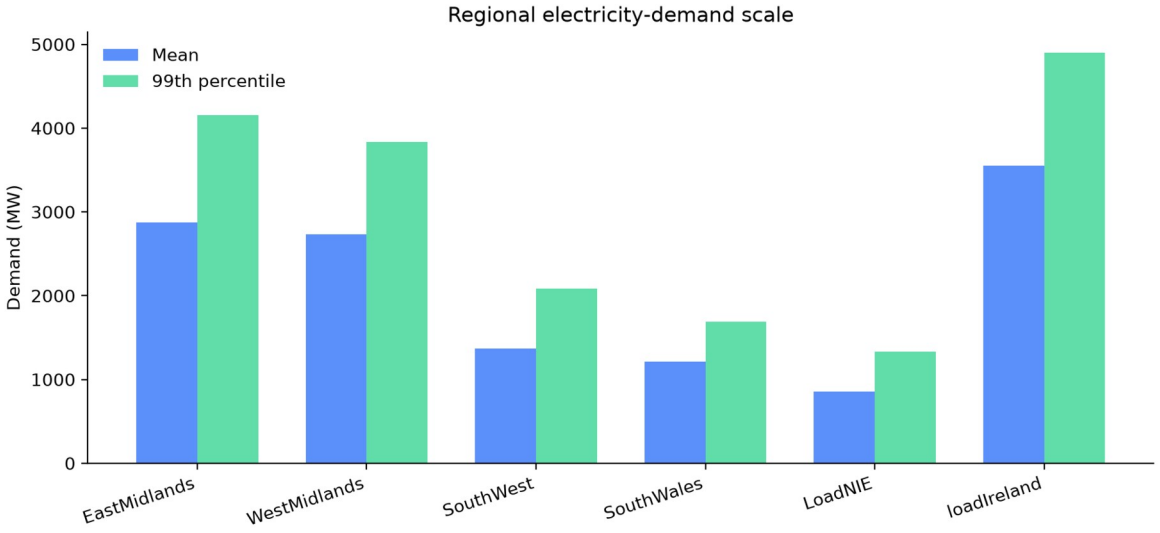


Figure 3.1. Regional electricity-demand scale

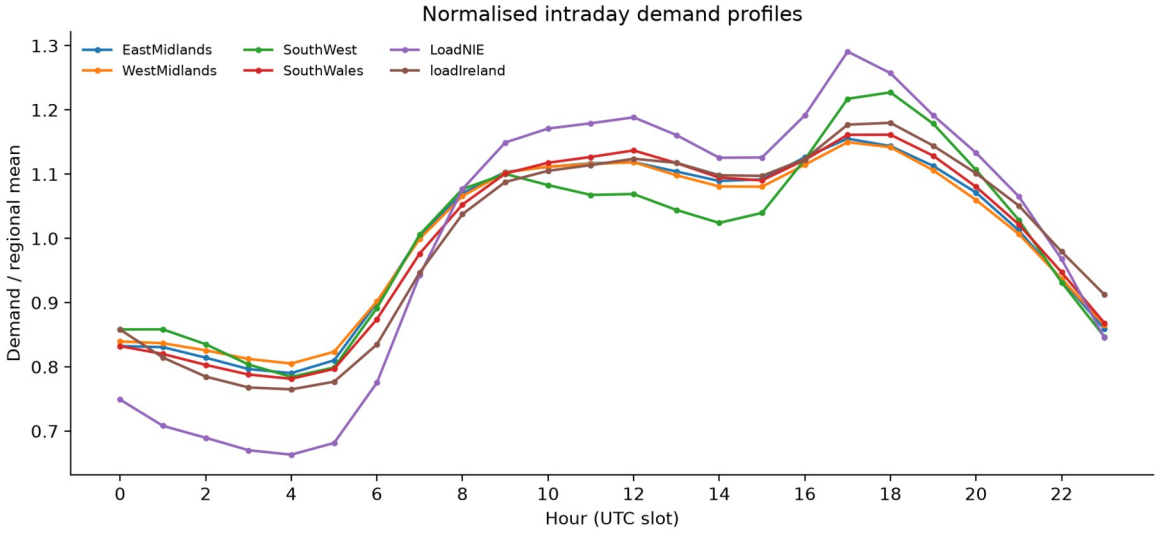


Figure 3.2. Normalised intraday demand profiles

Figure 3.3 adds the time dimension. It shows both recurring seasonality and changes in the average level over the 2020-2023 study window. These features motivate a chronological rather than random train-test split: random sampling would allow adjacent observations from the same seasonal regime to appear in both training and testing data and would overstate generalisation.

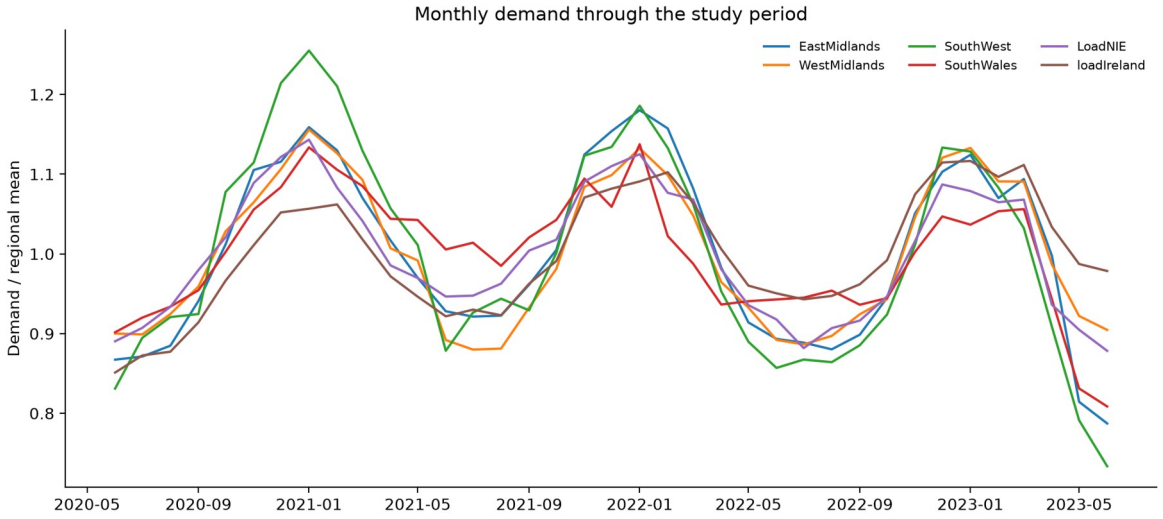


Figure 3.3. Monthly regional demand through the study period

Observations more than six median absolute deviations from the regional median are retained as auditable extremes because they may reflect genuine weather, holiday or social-event effects.

3.5 Economic and textual context

The term *text* refers to two distinct data products in this dissertation. The first consists of precomputed, numeric-valued conventional NLP features that are used as tabular predictors by the conventional forecasting models. The second is a separate catalogue of human-readable, source-linked event descriptions used by the LLM experiments. The two products are not interchangeable, and the event descriptions are not reconstructed from the precomputed NLP feature matrix.

The released text_features.csv contains 26,977 UTC-indexed hourly rows covering 1,125 dates from 1 June 2020 to 30 June 2023. All 1,044 predictor columns are numerical and contain no missing cells. Values are constant within each UTC day because the underlying news summary is daily even though the table is stored at hourly resolution. The columns combine corpus-size and text-length descriptors, word-frequency variables, 69 topic variables and 100 latent dimensions. The supplied hier_clusters_feat.pkl mapping assigns every one of the 1,044 columns to exactly one of ten clusters, whose membership sizes range from 24 to 369 variables.

The pipeline does not average raw columns directly. For each forecast horizon, every source variable is standardised using only news rows whose associated target date, forecast origin plus horizon, remains before 1 January 2022. Constant training variables receive a unit denominator. The ten model inputs are then formed as the row-wise means of the standardised variables in each supplied cluster and merged to predictors by forecast-origin timestamp. Cluster identifiers are retained as text_cluster_01 to text_cluster_10; they are not assigned post-hoc semantic topic names because the release provides heterogeneous memberships rather than ten validated human labels. Figure 3.4 shows that these leakage-safe aggregates retain non-zero temporal variation.

Economic context is provided in two files: the UK/NI file contains 3 numerical variables and the Ireland-specific file contains 3. These variables are joined at the forecast origin. The exploratory analysis reports their coverage and missingness but does not use full-period means for imputation, because that would leak validation and test information.

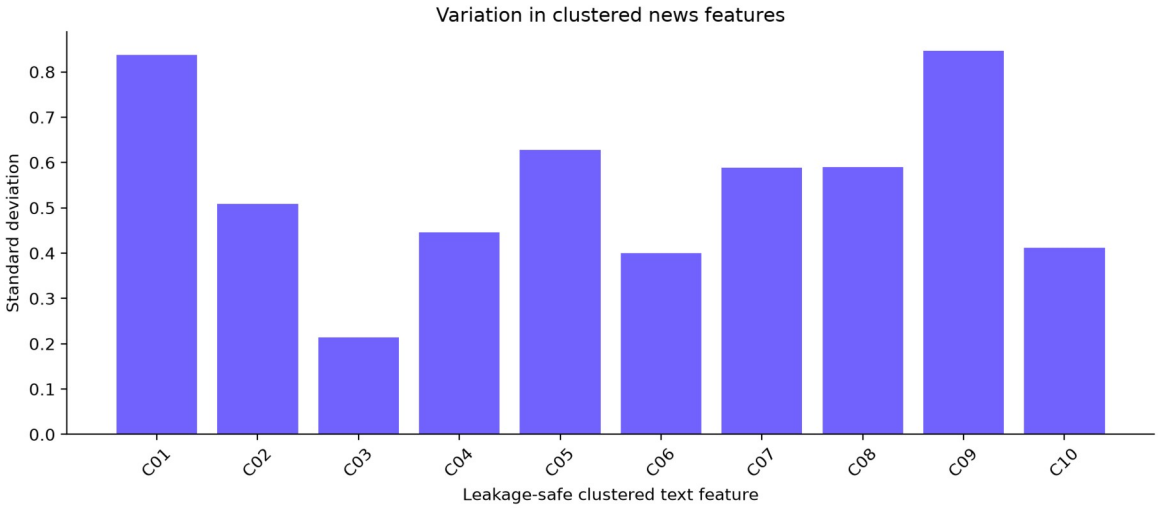


Figure 3.4. Variation in leakage-safe clustered text features

Explicit event descriptions used in the LLM experiments form a separate, source-linked table. Each record stores an event identifier, event dates, target geography, event type, short summary, verbatim source description, publisher, URL, publication time, operational availability time and archive or snapshot evidence. Analyst labels describing an expected direction or magnitude are audit fields only and are never supplied to the adjustment model. Publication and availability timestamps are checked against every forecast origin before the source text can be sent.

Table 3.2 makes the distinction between aggregated conventional NLP features and explicit event descriptions reproducible.

Table 3.2. Distinction between the two forms of textual information.

Dimension	Conventional clustered text features	Explicit event text
Source artifact	Published text_features.csv and hierarchical mapping	Separate frozen catalogue of official source descriptions
Observable representation	1,044 numerical variables; no released article corpus	Human-readable event summary and source text with URL
Time resolution	Hourly index with values constant within each UTC day	One record per dated event or event window
Reduction	Training-standardise each variable, then mean within 10 supplied clusters	LLM extracts a fixed schema; text is not embedded into the numerical baseline
Alignment	Joined at forecast origin; scaling boundary depends on horizon target date	available_at must be no later than forecast origin and dates must overlap the target
Model role	Candidate inputs to Ridge and LightGBM	Semantic routing: event type, dates, geography, demand channel and calibration key
Control over adjustment magnitude	Forecast coefficients are learned from chronological training data	Direct test asks for a bounded percentage; hybrid test moves magnitude to historical residuals
Main limitation	No raw articles or validated human label for each aggregate cluster	Small event catalogue and uneven support across event families

3.6 Implications for modelling

The EDA leads to five methodological decisions. First, the dataset represents each target date as a vector of 24 hourly demand observations, and this daily output structure is retained throughout the modelling pipeline. Second, forecast-origin and target-date fields remain separate. Third, preprocessing statistics use only data

available at the relevant modelling stage: training data for validation selection, followed by training and validation data for the frozen final test fit. Fourth, cross-region performance is reported using MAE in MW together with percentage-based metrics (MAPE, sMAPE and WAPE) and a Seasonal-Naive-relative MASE, so that differences in regional demand scale do not determine the comparison. Fifth, the defective source holiday indicators are replaced by jurisdiction-specific target-date calendar features. Together, these decisions ensure that economic and textual context is evaluated against a reproducible, leakage-safe conventional forecasting benchmark rather than against differences in regional scale or artefacts of alignment and preprocessing.

Chapter 4 Methodology

4.1 Research design

The staged chronological design benchmarks regional forecasts, tests explicit event-text adjustments, and then examines target-region exclusion, later years, non-calendar events, uncertainty, cost and an unseen electricity system. Earlier negative results are retained.

Figure 4.1 summarises the workflow. Structured tabular predictors produce the baseline; event text can change it only after availability, date, geography and support checks. Failed checks preserve the baseline, while passing events receive a bounded, historically calibrated correction rather than an LLM-selected magnitude.

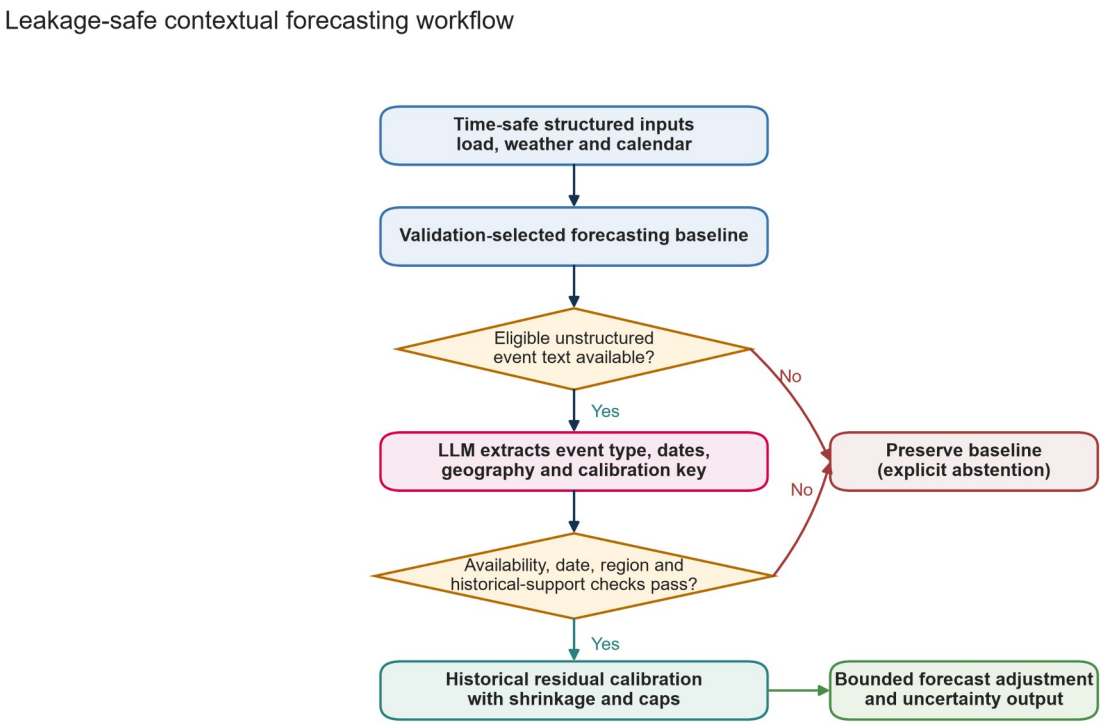


Figure 4.1. Leakage-safe contextual forecasting workflow

EXP012_R1 is the primary direct-LLM test; EXP013-EXP014 are post-hoc. The hybrid sequence is selected on 2022 before its paired 2023 test. Later-year national and French outcomes are untouched, while compatibility, robust-interval, global, strict regional and expanded-text analyses are labelled according to their inferential status in Appendix A.

Table 4.1 summarises the experimental groups and their inferential purpose.

Table 4.1. Experimental groups and inferential purpose.

Experiment group	Main purpose
EXP011-EXP021	Six-region baselines, controls, event calibration, ensemble and data audit
EXP022-EXP025	Later-year and baseline-compatibility transfer
EXP026-EXP029	Non-calendar events, abstention, intervals and cost
EXP030-EXP033	Global modelling, strict regional transfer, multi-fold text matching and external France test

Figure 4.2 shows the evidential sequence: benchmark, primary direct test, separately selected redesign and stress tests. Later mechanisms do not overwrite earlier negative or null findings.

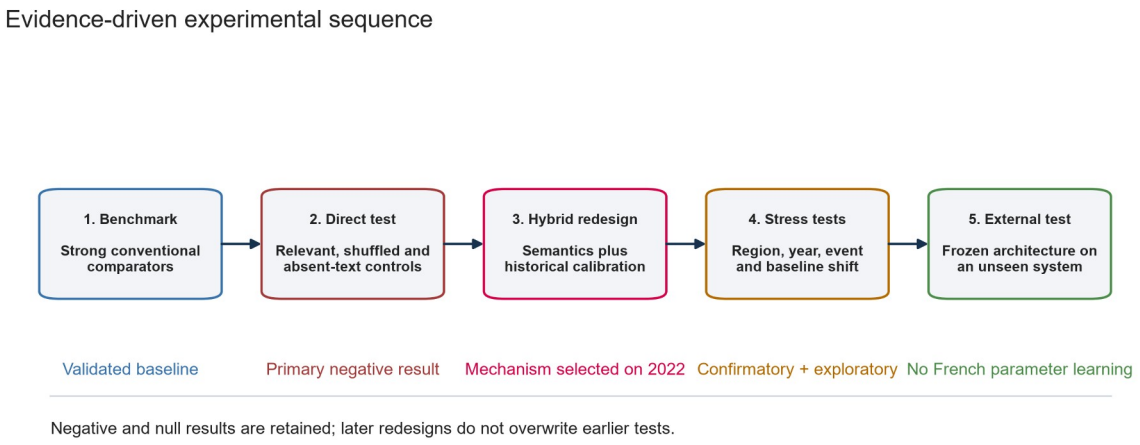


Figure 4.2. Evidence-driven experimental sequence

Operational definitions for forecast origin, target date, residual, semantic gate, historical calibration, partial pooling, abstention and target-region exclusion are consolidated in Appendix A.5.

4.2 Forecast task, splits and leakage control

The dataset represents each target date as a vector of 24 hourly demand observations. The forecast horizon determines the target day rather than the number of output hours: for region r, forecast origin t, horizon h from 1 to 30 and hour j from 0 to 23, each model predicts the complete 24-hour demand profile for date t + h. This daily output structure is retained throughout the modelling pipeline. Forecast origin and target date are stored separately:

$$\text{target date} = \text{forecast origin} + h \text{ days}$$

For each horizon, the target matrix Y therefore has 24 columns, with one column for each hour of the target date. A single training row contains the predictor values available at the forecast origin and the corresponding 24-hour target profile.

Economic and news variables must be observable at the origin. Calendar variables may describe the target because statutory dates are known in advance. Event text is eligible only when its recorded availability time is no later than the origin.

Training target dates precede 2022, validation covers 2022 and testing begins in 2023. Predictor retention, imputation, scaling and selection use training only; the frozen final model refits on training plus validation. Incomplete targets are excluded, while prediction artefacts retain fields required for metric recomputation.

4.3 Conventional predictors and models

In this dissertation, *conventional forecasting models* refers to Seasonal Naive, Ridge, LightGBM and their ensembles. The label *numerical condition* refers specifically to the predefined predictor set containing recent load, temperature and cyclical date variables; it does not denote a separate model class. Economic predictors are joined at the forecast origin, while the ten training-standardised clustered text features defined in Section 3.5 are added as conventional tabular inputs and are never sent to the LLM. Defective source holiday strings are replaced by a frozen target-date calendar for each jurisdiction.

Ridge and LightGBM are evaluated under four nested feature conditions: numerical; numerical plus economic; numerical plus economic and clustered text; and numerical plus economic, clustered text and the repaired calendar. Unsupported predictors are removed, missing values are replaced using fitting-partition medians, and Ridge scaling and target centring respect the same chronological boundary.

Seasonal Naive copies the latest observable complete same-weekday profile using lag $7 \times \mathrm{ceil}(h/7)$. Ridge is implemented in multi-response form. After training-only standardisation of the predictors and centring of each hourly target, the coefficient matrix is estimated as

$$B = (X^T X + \alpha I)^{-1} X^T Y$$

where Y contains the 24 hourly target columns. This matrix calculation is algebraically equivalent to fitting 24 hourly Ridge regressions with the same predictor preprocessing and regularisation parameter; it does not impose an additional dependence structure between the hourly outputs. The regularisation parameter α is selected from {0.1, 1, 10, 100}. LightGBM is instead fitted as 24 separate hourly models; its 16-point grid is selected on 2022 horizons 1, 15 and 30 and shared across horizons. Validation MAE selects the model-feature combination for each region-horizon cell.

The regional ensemble selects non-negative convex weights on a 0.1 grid. A scale-normalised global extension is enabled only when 2022 validation MAE improves by at least 2%; otherwise it falls back.

4.4 Direct LLM adjustment

The direct study uses six source-linked 2023 East Midlands clusters over horizons 1-7. Relevant-context, shuffled-context and no-context conditions contain the same 1,488 rows. At temperature zero, the gpt-4o-mini-2024-07-18 snapshot receives eligible source information, region, date, horizon and baseline mean, and returns direction, bounded percentage, duration, relevance, confidence and reason. Actual demand, forecast error and analyst labels are prohibited; requests and responses are retained.

$$\text{adjusted forecast} = \text{baseline forecast} \times \left(1 + \frac{a}{100}\right), \text{ with } -10\% \le a \le +10\%$$

The no-context condition defaults to zero. A predeclared deterministic rule and a Seasonal Naive symmetry check provide comparisons. The six clusters comprise five holiday groups and one cold alert, yielding 41 event-horizon observations; shuffling is frozen before inference.

4.5 Hybrid semantic gating and historical calibration

The second adjustment architecture is termed the hybrid framework because forecasting responsibility is divided among three components rather than assigned entirely to the LLM. The LLM performs semantic interpretation by extracting event type, dates, geographic applicability, demand channel, calibration key and confidence from the source text. A deterministic gate decides whether the extracted event information is eligible for use, while historical forecast residuals determine the direction and magnitude of any numerical adjustment. The LLM therefore has no direct authority to select an adjustment percentage.

Forecast-adjustment authority refers to responsibility for selecting the direction and magnitude of an adjustment and therefore determining the final forecast value.

The hybrid experiment evaluates this architecture on the same East Midlands event windows and underlying baseline used in the direct-adjustment experiment, allowing a paired comparison between the two allocations of forecast-adjustment authority. One LLM call is made for each event description rather than for every forecast horizon. The resulting structured record contains the event type, applicable dates, geographic scope, demand channel, calibration key, confidence and reason. The gate then checks whether the text was available at the forecast origin, whether the extracted dates overlap the target date, whether the geographic scope covers the target region and whether the calibration key is supported by historical evidence. Failure of any required check preserves the original baseline forecast.

Figure 4.3 isolates the design change: the direct route grants semantic and forecast-adjustment authority to the LLM, whereas the hybrid route gives adjustment authority to deterministic checks and observed residuals.

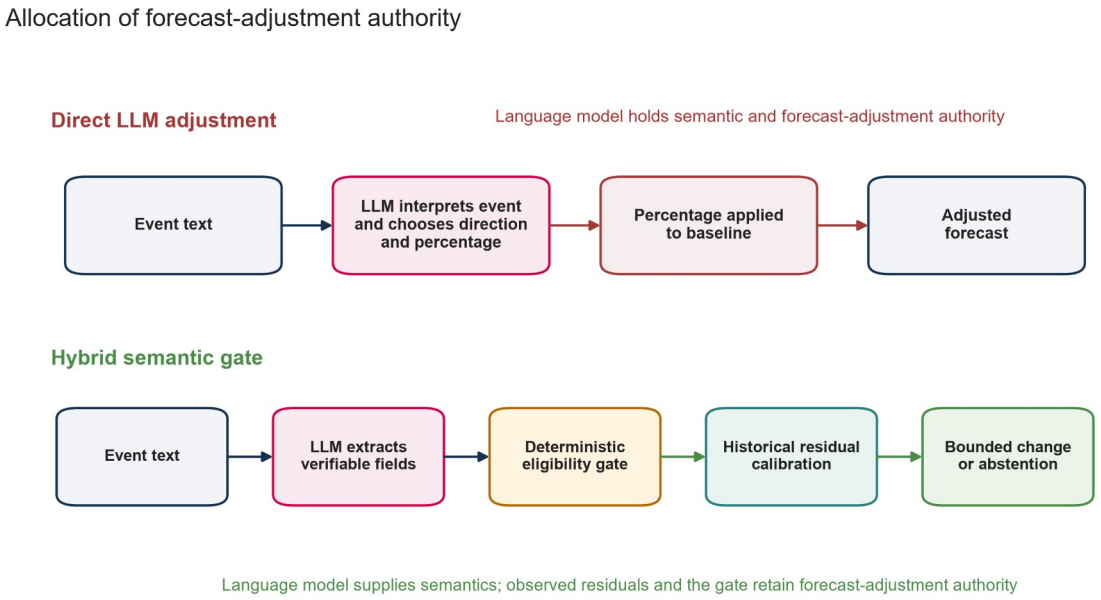


Figure 4.3. Direct and hybrid allocation of forecast-adjustment authority

Magnitude is learned from 2022 residuals using leave-one-event-out selection. Comparators include a capped Ridge residual specialist and a temperature-aware expert mixture. The hierarchical model partially pools regional and multi-region hourly factors:

$$\text{applied factor} = \text{clip}(\lambda \times \text{regional factor} + (1-\lambda) \times \text{pooled factor}, 1-c, 1+c)$$

The regional weight λ and cap c are frozen from 2022 leave-one-region-event-out validation. This layer uses language for applicability and historical data for direction and magnitude.

The Coronation bank holiday illustrates the complete hybrid route. The source text stated that the government had announced an additional bank holiday on 8 May 2023. The LLM converted this text into a fixed schema identifying a bank_holiday, the date 8 May 2023, East Midlands geographic applicability, a holiday_activity_change demand channel and the bank_holiday calibration key, with confidence 0.95. The deterministic gate then verified that the announcement was available before the forecast origin, that the extracted date overlapped the target date, that the geographic scope covered East Midlands and that the calibration key was supported. After these checks passed, hourly adjustment factors learned exclusively from 2022 bank-holiday residuals were applied to the baseline. The LLM did not observe the realised demand and did not select the direction or magnitude of the adjustment.

As sensitivity checks, a capped residual specialist uses hour, horizon, month, baseline level and event family, while an expert mixture includes holiday, temperature-aware cold and null routes.

4.6 Generalisation protocols

The strict held-out-region holiday test removes each target region's 2022 events from factor estimation. The full-year strict leave-one-region-out test is broader: the target region is removed from all model parameter learning for all 2023 target dates and all 30 horizons; target scaling uses only its already observed origin profile.

The later-year national test applies the frozen regional holiday factor to official NESO forecasts in 2024-2025 (National Energy System Operator, n.d.; GOV.UK, 2025). A post-hoc compatibility gate tests whether the operator baseline already accounts for holidays, and a subsequent guard is frozen on 2024 before available 2026 outcomes are downloaded.

The non-calendar test covers rail disruption, international football and cold weather, with target-region factors learned only from other regions' 2022 events. A route requires three independent events, two source regions, 2% leave-one-event improvement, bootstrap improvement probability of 0.80 and bounded uncertainty. The expanded catalogue contains 88 dated records and compares event-family pooling with term frequency-inverse document frequency (TF-IDF) neighbours across 2022 selection folds and a 2023 test; no adjustment is an explicit option.

For the external test, architecture is chosen from strict six-region leave-one-region-out anchors without French values. The frozen model predicts RTE's definitive 2024 load for all 366 days and horizons 1-30 (RTE, n.d.); 2023 French load is used only as time-safe history available at each origin.

4.7 Uncertainty, abstention and cost

This section evaluates three operational questions beyond point-forecast accuracy: how uncertainty around the forecast is quantified, when the hybrid system should abstain from adjustment, and how much LLM inference the two architectures require. These questions are examined separately because a model can improve point accuracy without providing calibrated uncertainty or a practical deployment mechanism.

EXP027 constructs nominal 90% relative-error conformal prediction intervals using only 2022 observations from eligible source regions; observations from the target region are excluded from interval calibration. For each calibration observation i, the relative conformity score is:

$$s_i = \frac{|y_i - \hat{y}_i|}{\max(|\hat{y}_i|, 1)}$$

With n calibration scores, q_{90} denotes the quantile selected at the finite-sample conformal rank. The interval around a forecast is:

$$[\max\{0, \hat{y}(1-q_{90})\}, \hat{y}(1+q_{90})]$$

The prediction interval follows the same routing decision as the point forecast. When the validation-frozen gate applies a hybrid adjustment, the interval is centred on the adjusted forecast and uses candidate conformity scores. When the gate abstains, the original baseline is preserved and the interval is constructed from baseline scores. The lower limit is truncated at zero because electricity demand cannot be negative.

These intervals target 90% coverage under the exchangeability assumptions required by conformal prediction (Angelopoulos and Bates, 2023). Because regional or electricity-system shift may violate those assumptions, coverage is evaluated explicitly rather than assumed to transfer. After under-coverage is observed, EXP029 replaces the pooled quantile with the maximum finite-sample source-region quantile. This modification is explicitly post-hoc and is therefore treated only as an exploratory robustness check, not as independent confirmation. The France interval audit is kept separate: its relative half-width is frozen using internal data before French outcomes are observed, and no French outcomes are used for calibration.

Finally, inference cost is evaluated independently of predictive accuracy. The audit compares one semantic extraction per event followed by local vectorised fan-out with repeated LLM calls for each event-horizon combination. Calls, tokens, recorded serial latency and price-at-audit estimates are compared on the same 1,488 forecast rows using prices recorded from the OpenAI model page on 10 August 2026. Historical price estimates and recorded network latency are therefore treated as audit measures rather than controlled billing benchmarks.

4.8 Evaluation metrics, statistical comparison and reproducibility

Let y_i and $\hat{y}_i$ denote observed and predicted demand for observation i, and let n be the number of evaluated observations. Mean absolute error is

$$\mathrm{MAE} = \frac{1}{n} \times \sum_{i=1}^{n} | y_i - \hat{y}_i |$$

and root mean squared error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \times \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

MAE is the principal metric because it remains in MW and weights errors linearly, while RMSE places greater weight on large errors. Percentage metrics support comparison across regions with different demand scales. Mean absolute percentage error is calculated over the m observations in the non-zero set I:

$$\mathrm{MAPE} = \frac{100}{m} \times \sum_{i \in I} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Symmetric MAPE and weighted absolute percentage error are

$$\mathrm{sMAPE} = \frac{100}{n} \times \sum_{i=1}^{n} \frac{2 | y_i - \hat{y}_i |}{(| y_i | + | \hat{y}_i |)}$$

$$\mathrm{WAPE} = 100 \times \frac{\sum_{i=1}^{n} | y_i - \hat{y}_i |}{\sum_{i=1}^{n} | y_i |}$$

MAPE is readily interpretable but unstable near zero; sMAPE reduces, but does not eliminate, denominator sensitivity; WAPE expresses total absolute error relative to total observed demand. All observed loads in the principal regional sample are positive.

For cross-region comparisons, the study reports a Seasonal-Naive-relative error ratio:

$$\mathrm{MASE}_{\mathrm{SN}} = \frac{\mathrm{MAE}_{\mathrm{candidate}}}{\mathrm{MAE}_{\mathrm{Seasonal\ Naive}}}$$

calculated on identical rows. Values below one favour the candidate. This is a study-specific relative MAE ratio rather than the textbook in-sample one-step MASE denominator; it is denoted MASE in the result tables for brevity.

Paired comparisons use

$$\Delta_{\mathrm{MAE}} = \mathrm{MAE}_{\mathrm{candidate}} - \mathrm{MAE}_{\mathrm{comparator}}$$

so negative values favour the candidate. Event experiments use a paired percentile bootstrap that resamples independent event clusters while retaining all dates, horizons and hours within each sampled event. Full-year and external-system experiments resample target dates while retaining their associated hours and horizons. The 2.5th and 97.5th percentiles form the reported 95% interval, and the improvement probability is the proportion of replicates for which Δ_{MAE} is below zero. Formal analyses use 5,000 replicates with fixed seeds; the supplementary EXP013-EXP014 analyses use 2,000. A result based on only one independent event cluster is treated as descriptive.

The bootstrap quantifies paired sampling variation but cannot compensate for a small number of independent events, post-hoc model selection or distribution shift. Protocols, input and code hashes, validation selections,

random seeds and point predictions are stored before formal outcome evaluation where applicable, allowing the reported metrics to be recomputed from frozen artefacts.

4.9 Research ethics and data governance

This study uses public secondary electricity-demand, weather, economic and event data. It does not involve human participants or process identifiable personal data. Explicit event descriptions are retained with their public sources and operational-availability timestamps, supporting data provenance and preventing information released after the forecast origin from entering the experiment.

LLM requests are restricted to the event information and forecasting fields required by the predefined schema. Realised demand, forecast errors and analyst-provided outcome labels are not supplied to the model. Requests and responses are retained for reproducibility, but API credentials are excluded from all stored research artefacts. LLM outputs are treated as experimental model inputs and are subject to deterministic availability, geographic and calibration checks rather than being accepted without validation. Appendix B contains the anonymised ethics-exemption evidence.

Chapter 5 Results

5.1 Conventional regional benchmark

The validation-selected benchmark completed six regions, 30 horizons, three model classes and four feature conditions. Table 5.1 reports the 2023 regional means. Every region has MASE below one, although South Wales is only marginally better than Seasonal Naive. Selection is heterogeneous: LightGBM is chosen at 93 horizons, Ridge at 77 and Seasonal Naive at 10; numerical-only features are selected most often, while clustered text is concentrated in Ireland. This supports a conditional rather than universal claim for conventional text.

Table 5.1. Regional 2023 conventional forecast performance.

Region	Selected MAE (MW)	Selected RMSE (MW)	Selected MAPE (%)	Selected sMAPE (%)	Selected MASE
East Midlands	194.21	243.89	7.55	7.16	0.88
West Midlands	92.65	123.06	3.28	3.28	0.57
South West	107.81	133.41	9.15	8.56	0.88
South Wales	82.91	105.05	7.39	7.08	0.96
Northern Ireland	44.16	58.40	5.44	5.26	0.65
Ireland	139.84	175.16	3.68	3.76	0.82

The repaired calendar improves 11 of 12 Ridge/LightGBM holiday comparisons and ties in one, with a mean 71.92 MW reduction. The validation-weighted ensemble improves the best-component comparison in four regions but transfers negatively in South West and Northern Ireland. Figure 5.1 shows the selected MASE pattern.

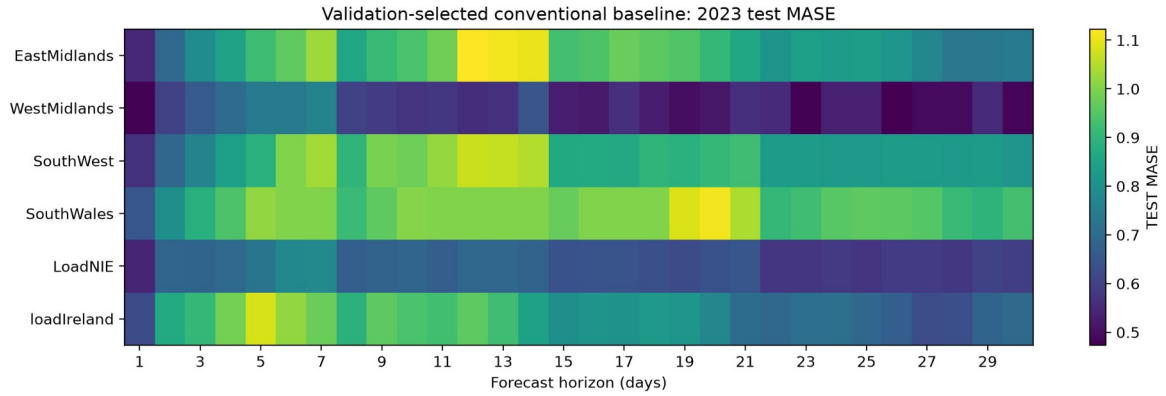


Figure 5.1. Validation-selected conventional MASE by region and horizon

5.2 Direct LLM failure and hybrid redesign

The frozen direct test evaluates identical East Midlands rows under relevant, shuffled and no context. LLM adjustment raises Ridge MAE from 239.80 to 310.23 MW (29.37%); the paired difference is +70.42 MW with 95% interval [21.74, 144.58]. Shuffled context records 290.64 MW, no context reproduces the baseline, and a deterministic rule reaches 197.96 MW. A Seasonal Naive check reaches the same conclusion, while repeated structured decisions show that reproducibility does not imply calibration.

Figure 5.2 diagnoses the frozen predictions using realised aggregate residuals that were neither supplied to the LLM nor used for retuning. Direction agrees in only 3 of 41 event-horizon cases: holidays generally

require a decrease but the LLM commonly proposes +5%, while cold-alert cases generally require an increase but often receive zero.

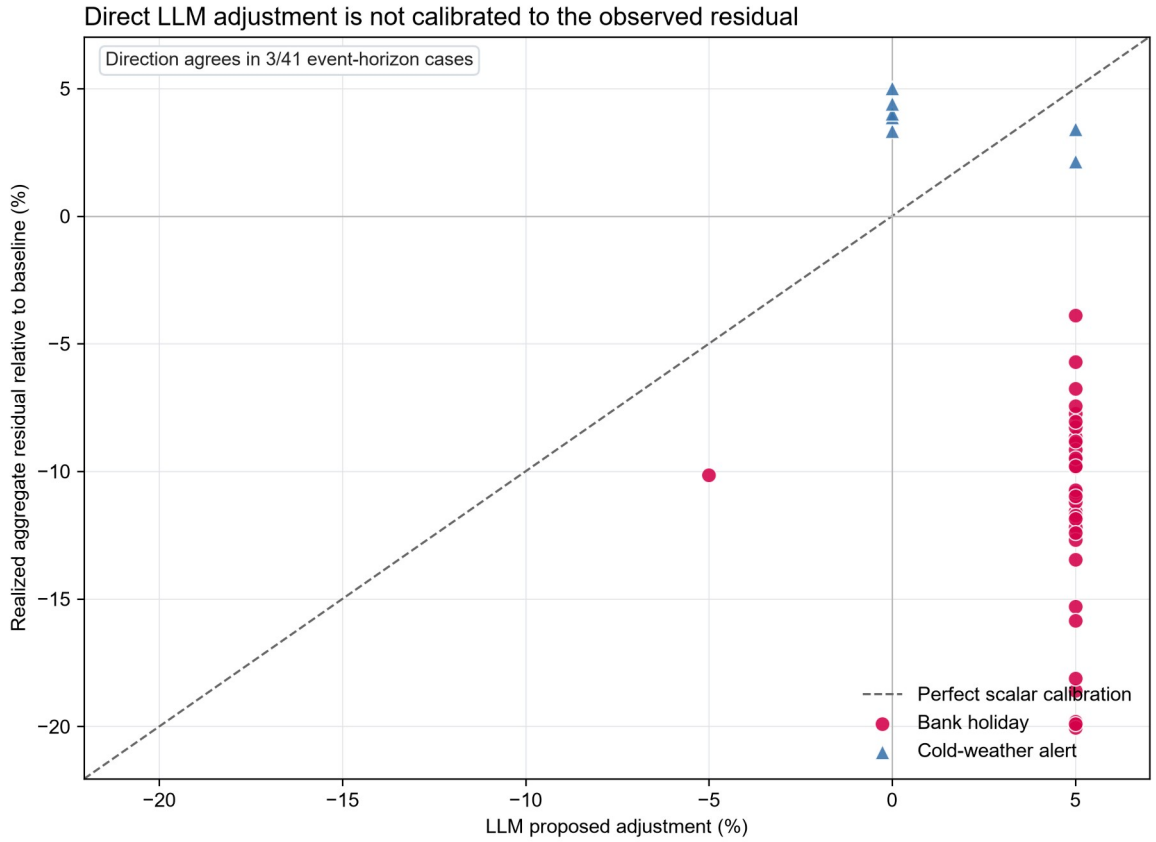


Figure 5.2. Direct LLM adjustment compared with the realised aggregate baseline residual

The hybrid changes responsibility rather than hiding the failure: the LLM extracts semantics and 2022 holidays determine the profile. MAE falls from 239.80 to 131.32 MW, difference -108.48 MW with interval [-220.11, -35.89]. Shuffled and no-context inputs preserve the baseline. Specialist variants give similar results; the single cold event remains descriptive. Table 5.2 summarises the comparisons.

Table 5.2. Direct and hybrid event-adjustment results.

Mechanism	Baseline MAE	Candidate MAE	Difference (MW)	Interpretation
Direct relevant LLM	239.80	310.23	+70.42	Confirmed deterioration
Semantic gate + calibration	239.80	131.32	-108.48	Confirmed event improvement
Holiday residual expert	287.11	131.86	-155.25	Confirmed on paired holidays
Holiday + cold experts	226.27	130.75	-95.52	Overall gain; cold evidence weak

For the Coronation example, the largest permitted hourly reduction occurred at 08:00, where the validation-learned factor of 0.80 changed the baseline forecast from 2,496.01 MW to 1,996.81 MW; the realised demand was 2,150.07 MW. Across all 168 Coronation event-horizon-hour observations, MAE decreased from 422.25 MW to 170.01 MW. This adjustment was produced by historical calibration rather than by an LLM-selected percentage. The cold-weather example demonstrates the opposite route: the LLM correctly identified the event type, dates, English geographic coverage and heating channel, but the text was unavailable at the relevant earlier forecast origins. The gate therefore applied a factor of 1.0 and preserved the baseline exactly.

Figure 5.3 decomposes the comparison by event. Direct LLM adjustment worsens all five holiday groups and improves the cold alert only slightly. The deterministic rule lowers holiday error without using language-model judgement, while historical calibration produces the largest holiday reductions and leaves the unsupported cold case at baseline. The plot is an error decomposition of the existing tests, not an additional independent experiment.

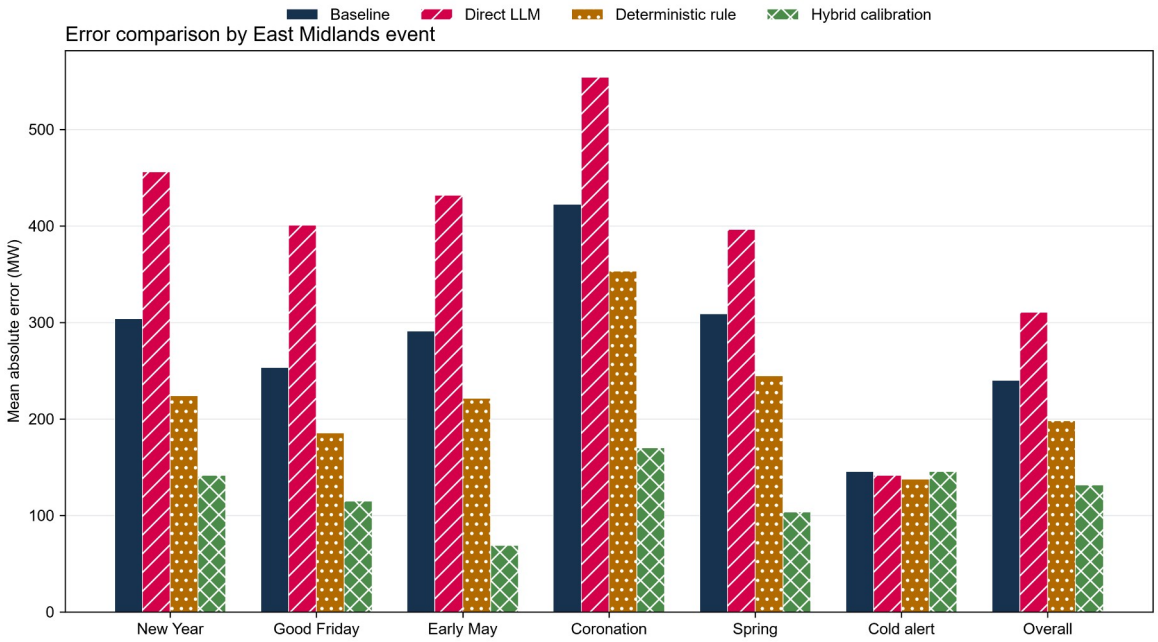


Figure 5.3. Baseline, direct, deterministic and hybrid MAE by East Midlands event

Six-region partial pooling selects a 0.75 regional weight and 20% cap. MAE falls from 159.76 to 81.36 MW across 35 region-event clusters, with difference -78.40 MW and interval [-109.78, -45.82]. Figure 5.4 shows improvement in every region; this validates holiday calibration rather than new regional LLM calls.

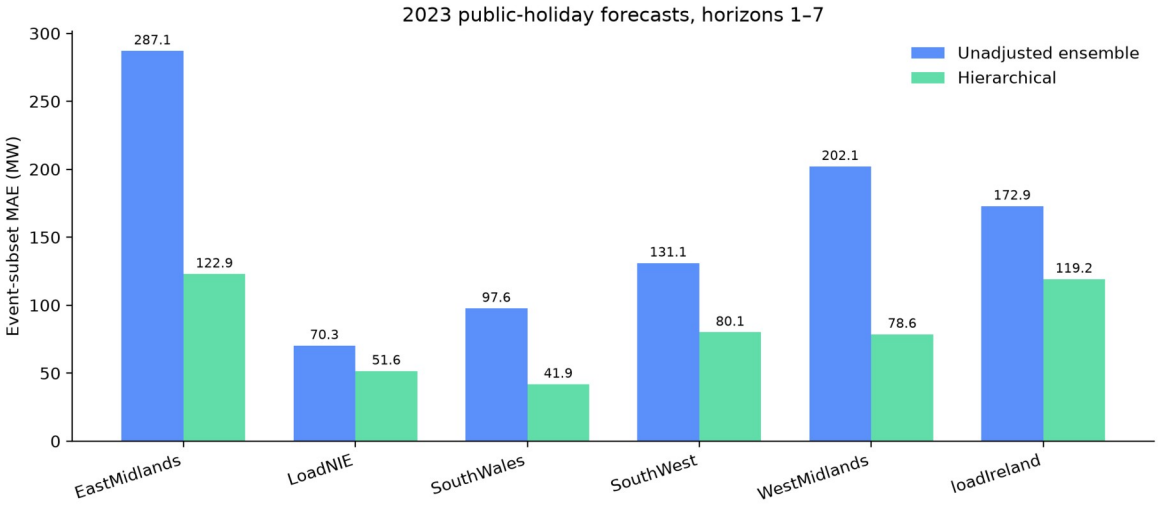


Figure 5.4. Public-holiday baseline and hierarchical MAE by region

5.3 Transfer across region, year and baseline

The strict held-out-region holiday test removes each target region's 2022 events from calibration. It lowers 2023 holiday MAE from 159.76 to 100.51 MW, a 37.09% improvement; the 95% interval for the difference is [-89.58, -27.66]. Five regions improve and Ireland worsens, indicating that transfer among British regional systems is easier than transfer across national-system and calendar boundaries.

Applying the regional factor to NESO's official 2024-2025 forecasts increases MAE from 1,055.31 to 1,870.26 MW, indicating double-adjustment of an event-aware baseline. A guard frozen on 2024 before available 2026 outcomes leaves candidate MAE at 1,085.73 versus 1,081.68 MW for NESO, supporting zero adjustment.

The retrospective global model enables 27 of 180 routes and reduces MAE from 105.02 to 103.55 MW (1.40%); most routes fall back.

The full-year strict leave-one-region-out test provides the stronger regional generalisation result. With the target region absent from all parameter learning, the 30-horizon global model reduces Seasonal Naive MAE from 142.87 to 123.34 MW, or 13.67%; the 95% interval is [-25.14, -14.36]. Four regions improve. Northern Ireland deteriorates 18.68% and Ireland 0.78%, preserving evidence of distribution mismatch.

Figure 5.5 shows improvements in the four British regions, while the two Irish-system series require safeguards.

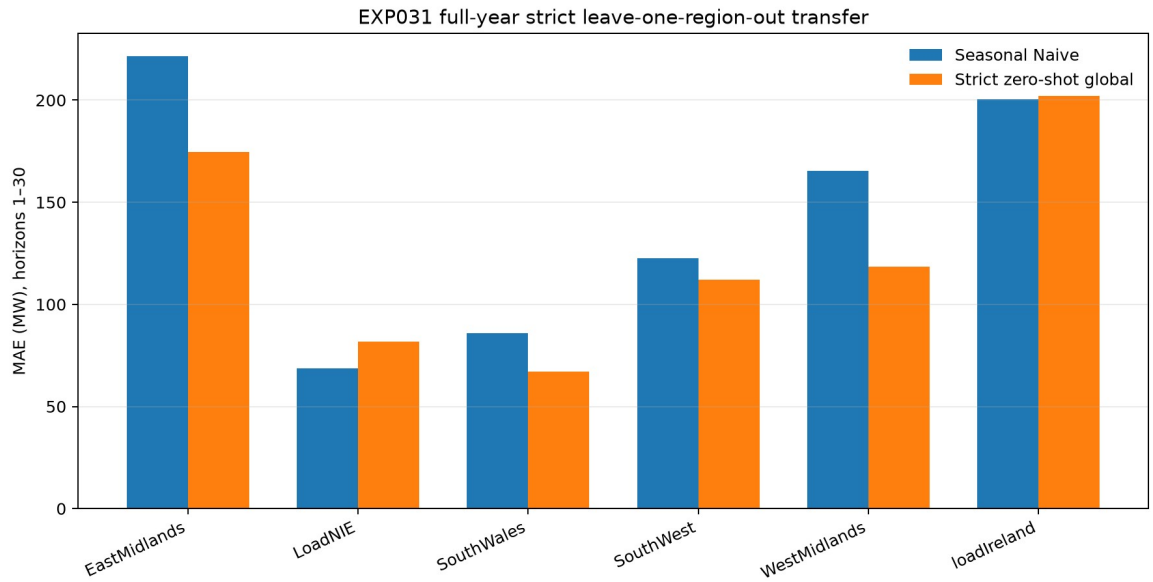


Figure 5.5. Strict leave-one-region-out full-year MAE

5.4 Non-calendar text, abstention and uncertainty

The non-calendar transfer test evaluates rail disruption, football and cold weather with target-region-excluded factors. Forced adjustment improves pooled MAE from 107.06 to 82.22 MW, but the event-cluster interval [-44.53, 8.80] crosses zero. Cold weather appears beneficial, rail gains are small and football worsens 13.00%. Because the cold calibration contains only one independent process, the frozen gate cannot treat its large point gain as sufficient evidence.

The gate authorises only the West Midlands rail route, adjusting 210 of 2,211 rows and abstaining on 2,001. MAE falls 0.70%, with an interval crossing zero. In the expanded catalogue, every pooled or TF-IDF option worsens late-2022 validation MAE by 1.68%-2.33%, so the frozen 2023 choice is no adjustment and test MAE remains 93.65 MW. Lexical similarity is insufficient to infer a stable event effect.

Figure 5.6 shows that the pooled nominal 90% interval covers only 83.45%. A post-hoc maximum source-region quantile raises coverage to 90.59%, but increases mean width from 371.38 to 448.06 MW. This is a robustness trade-off, not independent confirmation.

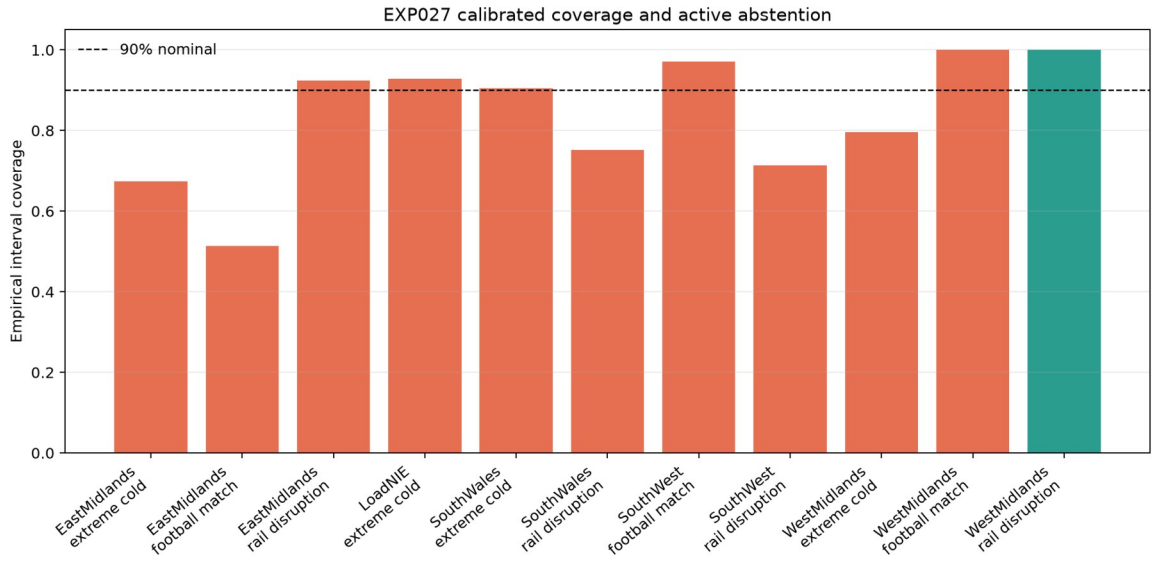


Figure 5.6. Accuracy, coverage and abstention for non-calendar events

5.5 Cost and unseen external system

For the same 1,488 event-hour forecasts, a per-horizon design makes 36 paid LLM calls and uses 14,904 tokens. One extraction per event makes six calls and uses 2,451 tokens. Figure 5.7 shows reductions of 83.33% in calls, 83.55% in tokens, 80.30% in the audited price estimate and 79.25% in recorded serial latency.

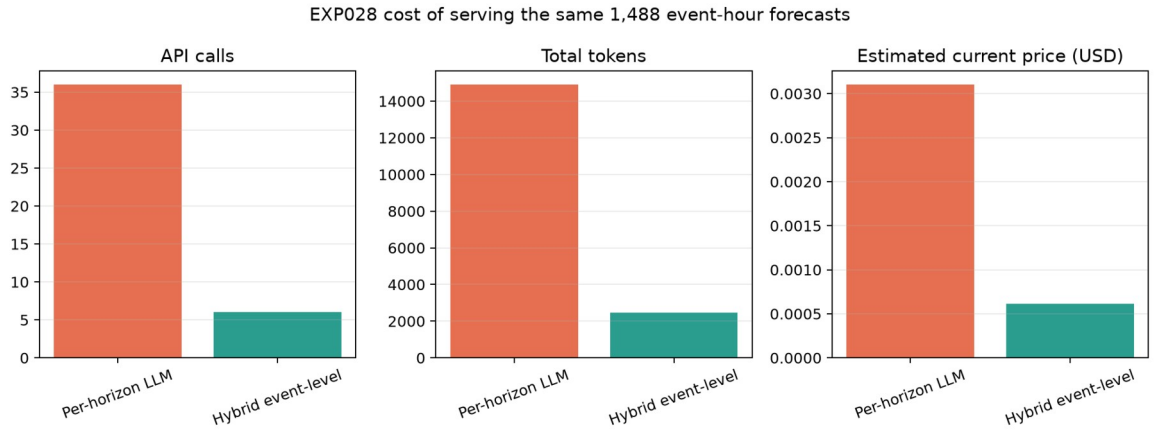


Figure 5.7. Hybrid and per-horizon LLM resource comparison

The France test is the external confirmation. France is absent from all parameter and architecture selection. On RTE's definitive 2024 load, the frozen scale-normalised Ridge model reduces Seasonal Naive MAE from 4,459.06 to 3,784.11 MW, a 15.14% improvement over 263,520 hourly predictions. The date-bootstrap difference is -674.95 MW with interval [-846.21, -509.07], and all 30 horizons improve. However, the frozen nominal 90% interval covers only 78.28%, so point-forecast transfer succeeds while uncertainty transfer fails.

Ridge with $\alpha = 100$ is selected from strict six-region anchors 1, 7, 14 and 30 before France is opened. Figure 5.8 shows improvement at every French horizon.

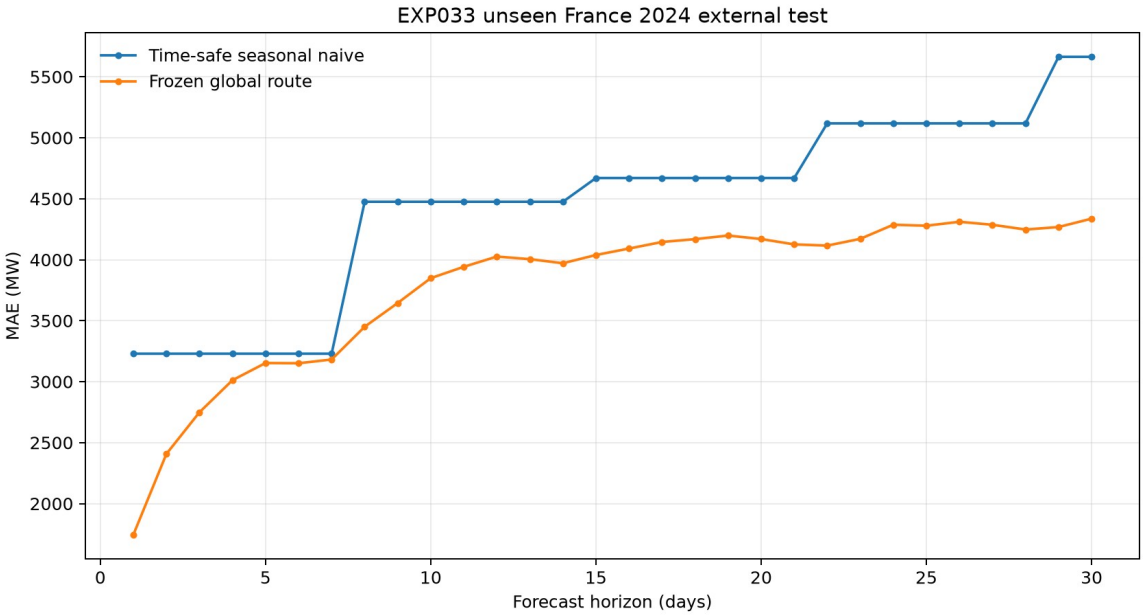


Figure 5.8. France 2024 external MAE by horizon

Table 5.3 consolidates the model families and prevents unlike datasets from being ranked by raw MW error.

Table 5.3. Summary of model families and their evidential role.

Model family	Evaluation scope	Paired result	Evidential interpretation
Seasonal Naive, Ridge, LightGBM and regional ensemble	Six regions, 2023, horizons 1-30	All regional MASE below 1; ensemble improves 4/6 regions	Credible but heterogeneous conventional benchmark
Direct LLM and deterministic rule	East Midlands event windows	239.80 to 310.23 MW for LLM; 197.96 MW for rule	Direct forecast-adjustment authority is rejected
Semantic gate, historical calibration and partial pooling	East Midlands and six-region holidays	239.80 to 131.32 MW; pooled 159.76 to 81.36 MW	Strong supported-event calibration; not proof an LLM is needed for known holidays
Held-out-region holiday and full-year regional transfer	Target region excluded	Holiday 159.76 to 100.51 MW; full year 142.87 to 123.34 MW	Positive transfer is conditional; Irish-system routes worsen
Non-calendar adjustment and abstention	Rail, football and cold events	Forced 107.06 to 82.22 MW; gated 106.31 MW	Forced point gain is uncertain; conservative routing is demonstrated
External France model	RTE France 2024, horizons 1-30	4,459.06 to 3,784.11 MW	Strong point transfer; nominal 90% interval covers only 78.28%

Results with different regions, dates or forecast units are not directly rankable by raw MAE. Each model is therefore shown against its own paired comparator.

5.6 Evidence strength and claim boundaries

The direct LLM adjustment evaluated in Section 4.4 increases MAE from 239.80 to 310.23 MW, a deterioration of 29.37%. By contrast, the hybrid semantic-gating and historical-calibration architecture defined in Section 4.5 reduces MAE from 239.80 to 131.32 MW, an improvement of 45.24%. Both estimates are paired event-window results calculated over 1,488 East Midlands observations; neither should be interpreted as a 45% reduction in annual regional forecasting error. The 13.67% strict regional-transfer improvement and the 15.14% France improvement instead arise from full-year, thirty-horizon evaluations.

These results support the transferability of scale-normalised point forecasting, but they do not independently establish the incremental value of event text.

Two experiments produce changes that are too small to support claims of material accuracy improvement. The retrospective global routing model reduces overall MAE by 1.40%. The non-calendar abstention route reduces MAE by only 0.70%, and its bootstrap interval includes zero. The principal value of these experiments is therefore methodological: they show how validation-based fallback and abstention can limit the use of an unsupported adjustment, rather than demonstrating a substantial reduction in forecast error.

The French result is evaluated separately against the Seasonal Naive forecast on the same French observations. Its absolute errors in MW are not compared directly with those of the six regional systems because their demand scales and evaluation settings differ. The relevant evidence is the paired reduction within the French system and whether that reduction is consistent across forecast horizons.

Appendix A.6 maps each headline result to the claim it supports and the claim it does not support. MAE reductions are forecasting outcomes rather than direct measures of reserve, dispatch or procurement value. No operational cost function is simulated, so economic-value claims are reserved for future work. Prediction-interval coverage remains separate because accurate point transfer does not guarantee calibrated risk under distribution shift.

5.7 Answers to the research questions

Conventional performance and contextual value are heterogeneous. Direct LLM numerical adjustment does not beat a strong baseline and is actively harmful in the primary test. The successful role is semantic: text can identify time, event family and geographic applicability, while historical models determine magnitude. Benefits are strongest for supported holidays; non-calendar effects require more independent events and frequent abstention. Strict regional and French external tests support scale-normalised point-forecast transfer, but negative regional routes and under-covered intervals prevent a universal deployment claim. Finally, event-level extraction is materially more practical than repeated horizon-level LLM calls.

Chapter 6 Discussion

6.1 Conditional answer to the research question

The evidence supports a conditional answer. A strong baseline is essential because load, weather and calendars already encode much context. Relevant event text is harmful when an LLM chooses the adjustment directly; improvement appears only when text determines applicability, historical residuals determine effect and a gate preserves the baseline under weak evidence. This refines the multimodal argument of Williams et al. (2025) and supports Tan et al.'s (2024) warning that access to context is not equivalent to calibrated numerical action.

6.2 Accuracy and conventional context

All regional means beat the time-safe seasonal benchmark, but accuracy and feature value remain heterogeneous. Clustered text is frequently selected for Ireland but rarely for East Midlands, consistent with Bai, Camal and Michiorri (2025), whereas the repaired target-date calendar improves nearly every holiday comparison. Known structured context should therefore be encoded before evaluating a semantic system.

6.3 Why direct adjustment failed

The direct LLM supplies coherent, repeatable fields but increases Ridge MAE from 239.80 to 310.23 MW, a 29.37% deterioration. The event-cluster interval [21.74, 144.58] excludes no effect, while relevant text does not clearly beat shuffled text. These controls indicate numerical miscalibration rather than malformed or unstable output.

Table 6.1 gives evidence-based explanations rather than a causal decomposition; the test cannot assign a unique share of error to each mechanism.

Table 6.1. Evidence-based explanation of the direct LLM failure.

Mechanism	Evidence in this dissertation	Interpretation and boundary
Direction and magnitude miscalibration	Structured fields repeat, but relevant adjustment increases MAE by 29.37%	Linguistic consistency is not numerical calibration
Missing counterfactual residual	The prompt contains text and a baseline mean, but no realised load or historical forecast errors	Knowing that an event affects activity does not reveal the residual relative to this baseline
Baseline information overlap	A later regional factor severely worsens an event-aware NESO forecast	An overlay can double-count context already absorbed by the baseline
Hour and horizon heterogeneity	One percentage is applied across an hourly profile, while event errors vary by hour and horizon	A scalar correction is too rigid for a changing load shape
Sparse and imbalanced event evidence	The primary test has six independent clusters: five holiday groups and one cold alert	The test supports a controlled failure claim, not universal event-effect estimation
Objective mismatch	The pretrained LLM is asked for a regional forecast correction without outcome-based fitting	General language knowledge supplies plausibility, not a locally estimated demand response

Event descriptions do not reveal a region-specific hourly counterfactual or what the baseline already captures. The deterministic rule and exact no-context fallback show why a zero-action state is safer than unsupported judgement. The direct approach is therefore rejected without rejecting semantic extraction when forecast-adjustment authority remains outcome-calibrated and abstention-aware.

6.4 Why the hybrid mechanism worked

The hybrid converts the LLM into an auditable parser while magnitude is learned from 2022 residuals. Similar factor, residual and partial-pooling results indicate that historical calibration, rather than one formula, drives the gain. Holiday results validate calibration more strongly than the need for an LLM because holidays are structured calendars. Non-calendar evidence is weaker: football transfers negatively and nearest-text matching selects no adjustment, supporting abstention rather than another post-hoc search.

6.5 What generalises - and what does not

Strict target exclusion and France provide the strongest generalisation evidence, with full-year gains of 13.67% and 15.14%. Transfer is nevertheless conditional: Northern Ireland and Ireland worsen, and a regional factor double-adjusts the event-aware NESO forecast. Uncertainty transfers less well, with nominal 90% coverage of 83.45% regionally and 78.28% in France; a wider post-hoc interval is not independent confirmation.

6.6 Practical architecture and cost

The recommended architecture encodes known calendars directly, structures time-safe event text once, checks availability, date, geography, support and baseline compatibility, and then applies a bounded residual model or preserves the baseline. Event-level extraction reduces calls and tokens by about 83%. Production would additionally require versioned sources, reliability rules and drift monitoring. Because procurement or reserve costs are not modelled, the audit establishes forecasting and computational practicality rather than monetary value.

6.7 Limitations and validity

The regional data cover June 2020 to mid-2023, and several extensions reuse 2023 outcomes. Events are few and holiday-dominated; the cold-weather result is descriptive and football transfer is negative. Aggregated news features prevent conventional NLP and LLM semantics from being compared on identical raw text. France adds only one national system-year and retains interval under-coverage. The direct result concerns one dated LLM snapshot and prompt, so it cannot represent every model. The holiday partial-pooling result is numerical transfer rather than new regional LLM generalisation, and the reported MASE is the time-safe study-specific ratio defined in Section 4.8. A future comparison should be preregistered on new event years and a shared corpus.

6.8 Overall contribution

The contribution is disciplined mechanism selection: LLMs can be useful semantic components when historical evidence and an explicit option not to act retain forecast-adjustment authority.

Chapter 7 Conclusion

7.1 Main findings

Whether event text improves short-term electricity-demand forecasts depends on the LLM's responsibility. Direct adjustment raises East Midlands MAE from 239.80 to 310.23 MW (29.37%): reproducible output remains miscalibrated.

A constrained hybrid succeeds on supported events: East Midlands MAE falls to 131.32 MW, and strict target-region exclusion improves pooled holiday MAE by 37.09%. The gains do not extend automatically: football transfers negatively, expanded text matching abstains and an event-aware NESO baseline is double-adjusted.

Full-year strict regional and unseen France tests improve Seasonal Naive by 13.67% and 15.14%, respectively. Yet nominal 90% coverage is only 78.28% in France. Event-level extraction reduces calls and tokens by about 83%.

7.2 Contributions and recommendation

The study contributes a leakage-safe benchmark, controlled negative result, semantic-gating redesign, transfer tests, abstention, interval audits and cost comparison. Its methodological conclusion is:

time-safe text -> semantic extraction -> deterministic eligibility gate -> historical calibration -> bounded correction or abstention

An LLM should not replace a conventional load forecaster or set an unconstrained percentage. It is justified only when unstructured text provides timing or applicability that structured variables do not already contain. Unsupported events, incompatible baselines and shifted systems should preserve the baseline or expose wider uncertainty.

Accordingly, the dissertation answers the main question with a qualified yes: textual context can improve forecasts through a constrained hybrid system, but direct LLM numerical judgement is not supported by the evidence. The external point result is retained without further tuning to the 2023 test set or to French 2024 outcomes.

7.3 Future work

The next study should freeze the full mechanism before a later event year, collect more independent non-calendar processes, and recalibrate intervals using data separate from the final target. Operational evaluation should translate forecast errors into reserve, dispatch or market cost. This would test whether the statistically careful hybrid architecture also produces decision value.

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Appendix A: Reproducibility and Evidence Provenance

This appendix makes the evidence chain understandable from the submitted PDF alone. Experiment identifiers are stable provenance keys; local folders and machine-specific file paths are deliberately omitted because they are not part of the assessed submission. The underlying code, configurations, hashes and machine-readable outputs are retained in the private project archive and can be supplied if formally requested.

A.1 Inferential status and decision rules

- **Frozen primary or external tests:** EXP012_R1, EXP016-EXP023, EXP025-EXP028 and EXP033, subject to the protocol in Chapter 4.
- **Retrospective extensions:** EXP030-EXP032 reuse 2023 outcomes and are labelled as such.
- **Explicit post-hoc robustness:** EXP013-EXP014, EXP024 and EXP029 do not redefine earlier tests.
- **Negative and null results are retained:** direct LLM numerical adjustment, NESO double-adjustment, most non-calendar routes, lexical matching and external interval under-coverage.

A.2 Experiment registry

The identifier records provenance, not evidential importance. Superseded development runs are retained for traceability but excluded from the dissertation's final numerical claims.

Experiment	Purpose	Status in dissertation
EXP001-EXP009	Early data, baseline and LLM-adjustment feasibility iterations	Superseded development; excluded from final numerical claims
EXP010	Deterministic event-adjustment pipeline and time-safety demonstration	Prototype only
EXP011	East Midlands chronological Seasonal Naive, Ridge and LightGBM benchmark	Formal conventional benchmark
EXP012_R1	Relevant, shuffled and absent-context direct LLM test	Frozen primary negative result
EXP013-EXP014	Direct-output sensitivity, failure mechanism, baseline symmetry and prompt stability	Post-hoc robustness; does not redefine EXP012_R1
EXP015_R1	Six-region conventional models, feature ablations and repaired calendars	Formal benchmark extension
EXP016	LLM semantic extraction plus validation-selected historical calibration	Frozen hybrid mechanism test
EXP017-EXP018	Residual specialist and holiday/cold expert mixture	Frozen numerical mechanism comparisons
EXP019	Validation-weighted Seasonal Naive, Ridge and LightGBM ensemble	Frozen working baseline
EXP020	Partially pooled six-region holiday calibration	Frozen cross-region calibration
EXP021	Data integrity and exploratory analysis	Descriptive audit
EXP022	Frozen holiday-factor transfer to NESO 2024-2025	Negative external baseline-transfer test
EXP023	Strict target-region-excluded holiday transfer	Frozen positive regional transfer test
EXP024	Baseline-compatibility gate after the NESO failure	Explicitly post-hoc
EXP025	Guard frozen on 2024 before available 2026 outcomes	Small-sample prospective check
EXP026	Target-excluded non-calendar event transfer	Frozen candidate test; interval includes no effect
EXP027	Evidence gate, active abstention and nominal intervals	Frozen conservative-routing test
EXP028	Event-level extraction versus per-horizon LLM cost	Audited operating-cost comparison
EXP029	Maximum source-region conformal quantile	Post-hoc interval robustness
EXP030	Validation-gated scale-normalised global model	Retrospective 2023 extension
EXP031	Full-year strict leave-one-region-out model	Retrospective but target-excluded transfer test
EXP032	Expanded catalogue and two-fold text matching	Retrospective null-selection result
EXP033	Frozen architecture applied to unseen RTE France 2024	External point-prediction confirmation; interval failure retained

A.3 Reproducibility information retained

Each formal run retains its data snapshot or source dates, event availability times, split definition, random seed, model and prompt schema, configuration, code and input hashes, point predictions, intervals and recomputable metrics. The non-calendar catalogue also records publisher, source URL, event dates, geographic scope, action hours and publication time. These records are keyed by the experiment IDs above; no local path is required to interpret the thesis.

A.4 LLM request and audit fields

The direct schema stores direction, bounded percentage, duration, geographic relevance, confidence and reason. The hybrid schema stores event type, dates, region scope, demand channel, calibration key, confidence and reason, but contains no direction or percentage field. For each logical request the archive retains the model snapshot, messages, schema, prompt hash, raw structured response and parsed checkpoint. Actual demand, forecast error and analyst expected-direction fields are excluded from prompts.

A.5 Operational glossary

Term	Definition in this dissertation
Forecast origin	Timestamp at which the prediction and every admissible input are assumed available
Target date	Date being predicted; equal to forecast origin plus horizon
Baseline	Forecast produced before any explicit event-text overlay
Residual	Actual demand minus the corresponding baseline forecast
Semantic gate	LLM extraction followed by deterministic availability, date, geography and support checks
Historical calibration	Adjustment direction and magnitude estimated from earlier forecast residuals
Partial pooling	Weighted combination of regional and multi-region event-effect estimates
Abstention	Exact preservation of the baseline when eligibility or evidence requirements fail
Target-region exclusion	No target-region outcomes are used to learn model parameters or event factors

A.6 Evidence-to-claim boundaries

The table states exactly what each result can and cannot justify; it is the main reason Appendix A is retained in the PDF.

Finding	Evidence strength	Supports	Does not support
Conventional benchmark beats Seasonal Naive on regional means	Full 2023; six regions; 30 horizons	Baseline credibility and regional heterogeneity	Uniform superiority at every region-horizon cell
Direct LLM adjustment is harmful	Frozen paired test; event-cluster interval excludes zero	Failure of the tested direct mechanism	Failure of every LLM, prompt or future model
Hybrid calibration improves supported events	Paired East Midlands test plus six-region holiday consistency	Separation of semantic and numerical responsibility	That an LLM is necessary for known calendar holidays
Strict regional and French point transfer improves	Target-region exclusion and unseen France system-year	Transferability of scale-normalised point forecasts	Universal transfer or calibrated external intervals
Non-calendar gate mostly abstains	Frozen thresholds and negative selection result	Safe preservation of the baseline under weak evidence	Material non-calendar accuracy improvement
Event-level extraction reduces inference resources	Same served rows; audited calls, tokens and recorded latency	Practical advantage over per-horizon LLM calls	A timeless provider-price or network-latency benchmark

Appendix B: Ethics Exemption and Data Protection Evidence

This project uses public, anonymised secondary electricity-demand, temperature, economic, calendar, news-feature and event-source data. It involves no living participants, interviews, surveys, private records or identifiable personal data. No decisions are made about individuals. API requests contain whitelisted event fields and exclude actual target outcomes and API credentials. On 17 June 2026, the project route was confirmed with the supervisor as exempt from research ethics review on this basis.

B.1 Anonymised Step A6 declaration

The following is the anonymised Step A6 statement from the completed MSc ESDA Research Ethics, Data Protection and Risk Check. Personal names and email addresses are omitted for anonymous second marking.

Statement of Risk Assessment & Ethics Approval Requirements
Student Candidate Number: WGJB1 Student Name: Student UCL Email Address: **Supervisor Name:** Akylas Stratigakos Supervisor UCL Email Address: a.stratigakos@ucl.ac.uk **Dissertation Research Proposal**: • Title / Topic: Using textual information and Large Language Models to improve short-term electricity demand forecasting. • Research Question(s) / Aims & Objectives: Can textual information help improve electricity demand forecasts? Can an LLM use event-related text to adjust forecasts more accurately and reliably? • Data & source (specify all data to be used; if none, explain why): Publicly available regional electricity demand, weather, calendar, economic and news data for the UK and Ireland, together with event-related textual information. • Method(s) (specify all methods to be used): Time-series forecasting, machine learning (Seasonal Naive, Ridge Regression and LightGBM), and LLM-based textual analysis and forecast adjustment.
I have read and understood **Step A1 'Does the research require a Risk Assessment?'** and this planned research does NOT require a risk assessment.
I have read and understood **Step A2 'Does the research require External research ethics approval?'** and this planned research does NOT require external ethics review.
External ethics approval is *not required* and I have read and understood **Step A3 'Is the research Exempt from the need for ethics approval?'** and this planned research IS EXEMPT from the need for research ethics approval.
I confirm that: • the information I have provided is accurate to the best of my knowledge. • if the answers to any of these questions changes, I will go through this protocol again.

The final study remained within this approved route: it used public secondary system-level data and public event announcements, with no participants, private records or identifiable personal data.